Numerical Solution of a Variable-Diffusion Elliptic BVP

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Abstract

I solve the steady variable-diffusion problem $-\nabla \cdot (a\nabla u) = f$ on the unit square (Problem Pack Direction 3) by a conservative face-averaged five-point finite-difference method, assembled in COO and solved as CSR. Dirichlet data are eliminated into the right-hand side; boundary nodes are never unknowns. On the smooth manufactured benchmark $a = 1 + x$, $u^* = \sin(\pi x)\sin(\pi y)$, the discrete L^2 and L^∞ errors both fit order 2.00 over $n = 15, 31, 63, 127$, and $\|A - A^\top\|_{\max}/\|A\|_{\max} = 0$. On the layered jump $a = 1\,|\,10$ at $x = 1/2$, a harmonic (resistance) face average reproduces the piecewise-linear exact solution at the nodes to 10^{-14} and keeps the interface flux at $20/11$ to the linear-solve accuracy; an arithmetic average of the same two nodal values yields only first-order L^∞ errors and a flux jump that does not vanish under refinement. Unpreconditioned CG, stopped on the residual, matches the direct L^2 error once the relative residual is 10^{-4}; at $n = 31$ there is no runtime crossover with `spsolve`. The main limitation is that the scheme is first-order at an unresolved jump if the face coefficient is chosen arithmetically, and that CG iteration counts grow with both $1/h$ and the contrast a_R/a_L.

Keywords: elliptic BVP, variable diffusion, five-point stencil, harmonic average, method of manufactured solutions, sparse direct / CG

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1 Introduction and Motivation

1.1 Problem Background

Steady heat conduction in a heterogeneous medium—a layered composite, a soil profile, a tissue boundary—is the model problem in which the conductivity $a(x, y)$ is not constant and may jump [8, 1]. The physically correct operator is the divergence form $-\nabla\cdot(a\nabla u)$, because that is the statement that the normal flux $a\partial_n u$ is continuous across an interface. A pointwise expansion $a\Delta u+\nabla a\cdot\nabla u$ is the wrong starting point once a is discontinuous: it loses conservation, symmetry, and the M-matrix sign pattern.

I chose Direction 3 rather than a curved-disk Poisson problem because the unit square lets the five-point machinery from Tutorial 5 stay intact, while the jump forces a genuine modelling choice (arithmetic versus harmonic face coefficients) that a smooth manufactured Poisson problem never sees. This report is a solo submission: I derived the stencil, wrote the solvers, ran the experiments, and typeset the text.

1.2 Mathematical Model

On $\Omega = (0, 1)^2$ with Dirichlet data on the whole boundary,

$$-\nabla\cdot\big(a(x,y)\nabla u\big) = f \quad \text{in } \Omega, \qquad u = g \quad \text{on } \partial\Omega, \tag{1}$$

with $0 < a_{\min} \le a(x, y) \le a_{\max} < \infty$. The sign convention is the one that makes $-\nabla\cdot(a\nabla u)$ a positive operator: the discrete matrix will have a positive diagonal.

Smooth manufactured benchmark (bank P4).

$$a(x,y) = 1 + x, \qquad u^*(x,y) = \sin(\pi x)\sin(\pi y), \qquad g = 0,$$

and

$$f = 2\pi^2(1 + x)\sin(\pi x)\sin(\pi y) - \pi\cos(\pi x)\sin(\pi y). \tag{2}$$

The extra $-\pi\cos(\pi x)\sin(\pi y)$ term is $\nabla a\cdot\nabla u^*$; it is the reason a constant-coefficient five-point stencil is not enough.

Layered-interface benchmark.

$$a(x) = \begin{cases} 1, & x < 1/2, \\ 10, & x > 1/2, \end{cases} \qquad f = 0,$$

and $g = u^*$ on $\partial\Omega$, where the exact one-dimensional profile

$$u^*(x) = \begin{cases} \dfrac{20}{11}\, x, & 0 \le x \le 1/2, \\ \dfrac{2}{11}\, x + \dfrac{9}{11}, & 1/2 \le x \le 1, \end{cases} \tag{3}$$

is independent of y. It is continuous, $u^*(0) = 0$, $u^*(1) = 1$, the derivative jumps, and the flux $au^*_x = 20/11$ is continuous. Well-posedness follows from uniform ellipticity and a nonempty Dirichlet boundary; no compatibility condition is required.

Expected signs: $u^* \ge 0$ on the layered problem, with extrema 0 and 1 attained on the boundary (maximum principle). The smooth P4 solution is a positive first eigenfunction of $-\Delta$ modulated by $a = 1 + x$, so $u^* > 0$ in Ω and $u^* = 0$ on $\partial\Omega$.

1.3 Overview of the Report

Section 2 derives the face-averaged stencil, the Dirichlet shift, truncation, and the discrete structure (symmetry, SPD, M-matrix). Section 3 records the implementation. Section 4 verifies P4 by MMS, separates algebraic from discretisation error, compares harmonic and arithmetic averages on the jump, and reports cost. Section 5 states what I trust.

2 Discretisation and Analysis

2.1 Discrete Scheme

I use a uniform nodal grid with n interior nodes per direction,

$$h = \frac{1}{n+1}, \qquad x_i = (i+1)h, \qquad y_j = (j+1)h, \qquad i, j = 0, \dots, n-1. \tag{4}$$

The unknown vector is indexed by $k = in + j$ (`meshgrid indexing='ij'`, axis 0 is x). Boundary nodes are not unknowns.

A conservative finite-volume / face-centred reading of $-\nabla \cdot (a\nabla u)$ on the dual cell of width h gives the five-point divergence-form stencil [2, 8]

$$(A_h U)_{ij} = \frac{1}{h^2}\Big[a_E(U_{ij} - U_{i+1,j}) + a_W(U_{ij} - U_{i-1,j}) + a_N(U_{ij} - U_{i,j+1}) + a_S(U_{ij} - U_{i,j-1})\Big]. \tag{5}$$

When $a \equiv 1$ this is the classical five-point Laplacian

$$(-\Delta_h u)_{ij} = \frac{4u_{ij} - u_W - u_E - u_S - u_N}{h^2}.$$

The four face coefficients are shared: the east coefficient of node (i, j) is the west coefficient of node $(i + 1, j)$. That sharing, not the particular average, is what makes A_h symmetric after Dirichlet elimination.

For smooth a I evaluate the coefficient at the half-point, $a_E = a(x_i + h/2, y_j)$. For $a = 1 + x$ this coincides with the arithmetic nodal midpoint and is second-order. At a material interface I use

the resistance (harmonic) average of the two nodal values,

$$a_E = \frac{2a_L a_R}{a_L + a_R}, \tag{6}$$

and, as a deliberate comparison, the arithmetic mean $(a_L + a_R)/2$. When the interface bisects a grid edge, (6) is exact for a piecewise-constant a. I therefore take odd interval counts $N = n + 1 \in \{31, 63, 127, 255\}$ on the layered test, so $x = 1/2$ lies midway between two interior columns [8].

2.2 Consistency and Truncation Error

Taylor expansion of (5) for smooth a and smooth u reproduces $-\nabla \cdot (a\nabla u)$ with leading term $\mathcal{O}(h^2)$ (the same u_{xxxx} and u_{yyyy} pieces as the five-point Laplacian, plus smooth lower-order terms from the variation of a). The assumption is $u \in C^4$ and $a \in C^2$. It is false across the jump: the layered u^* is only Lipschitz, and I do not claim second order for an arithmetic face coefficient there.

I measure the truncation residual $\tau_h = b - A_h u^*$ in the discrete L^2 norm $h\|\tau_h\|_2$. On P4 this quantity drops by four under each mesh doubling (Table 2), consistent with $\mathcal{O}(h^2)$ pointwise truncation.

2.3 Boundary Treatment

Every neighbour that lands on $\partial\Omega$ is removed from the unknown list. If the stencil at interior index k has a Dirichlet neighbour with value g_{edge}, the term $a_{\text{face}}(U_k - g_{\text{edge}})/h^2$ contributes

$$b_k \mathrel{+}= \frac{a_{\text{face}}\, g_{\text{edge}}}{h^2} \tag{7}$$

to the right-hand side, with the same sign as moving $-(-a_{\text{face}}g_{\text{edge}})/h^2$ across. For P5 ($u^* = e^x \sin(\pi y)$, $a = 1$) the west column $i = 0$ is the row set $k = 0, \ldots, n-1$; an example is

$$b[0] \mathrel{+}= \frac{e^0 \sin(\pi y_0)}{h^2}.$$

Replacing a Dirichlet row by e_k while leaving the corresponding column untouched would destroy symmetry [8]. I never do that: boundary nodes are absent from A_h.

This project is pure Dirichlet. There is no Neumann/Robin closure and therefore no one-sided versus ghost-point comparison.

2.4 Discrete Structure

After Dirichlet elimination, every off-diagonal entry of A_h is non-positive and the row sums on interior stencils that do not touch the boundary are zero; boundary-adjacent rows are strictly diagonally dominant because a positive a_{face}/h^2 has been moved to b. With positive shared face coefficients, A_h is a symmetric M-matrix, hence SPD, and the discrete maximum principle holds: a discrete solution of $A_h U = 0$ with Dirichlet data in $[m, M]$ remains in $[m, M]$.

I check three numerical corroborations, none of which replaces the derivation: $\|A - A^\top\|_{\max}/\|A\|_{\max} = 0$ on every assembled matrix in this report; the smallest algebraic eigenvalue of the P4 matrix at $n = 15$ is $\lambda_{\min} = 28.61 > 0$; the layered numerical solution stays in $[0, 1]$. Conditioning is a separate fact: $\kappa_2(A_h)$ on P4 grows like h^{-2} (126.5, 529.7, 2.18×10^3 at $n = 15, 31, 63$), as expected for a second-order elliptic operator.

3 Implementation

3.1 Algorithm

Algorithm 1 Assembly of $A_h U = b$ and the direct solve

Require: interior count n, coefficient a, data f, g, face rule (half-point / harmonic / arithmetic)
Ensure: u_h, residual $\|b - A_h u_h\|_\infty$

$h \leftarrow 1/(n+1)$; index $k = in + j$
$b \leftarrow f$ sampled at interior nodes
for each interior node (i, j) **do**
    for each of the four faces **do**
        compute a shared $a_{\text{face}} > 0$
        if the neighbour is interior **then**
            add a_{face}/h^2 to the diagonal and $-a_{\text{face}}/h^2$ to that column
        else
            add a_{face}/h^2 to the diagonal and $a_{\text{face}} g_{\text{edge}}/h^2$ to b_k
        end if
    end for
end for
convert COO triplets to CSR; $u_h \leftarrow$ `spsolve`(A, b)
report $\|b - Au_h\|_\infty$

3.2 Code Structure

File	Purpose
`model.py`	$a(x, y)$, manufactured u^*, layered exact profile (3)
`assemble.py`	sparse face-averaged stencil, index map, interface flux
`solvers.py`	`spsolve`, residual-stopped CG, $\lambda_{\min}$ / κ via `eigsh`
`verify.py`	named-norm errors, four-mesh tables, layered diagnostics
`plots.py`	every report figure
`fem1d.py` / `fem2d_oracle.py`	independent 1D P1 and optional 2D P1 oracles
`test_problem_bank.py`	analytic bank (P1–P5), self-check of $f = -\nabla \cdot (a \nabla u^*)$
`validation_tests.py`	sign, boundary, order, nodal-exact, and M-matrix checks
`run_all.py`	regenerates every figure and prints the quoted numbers

3.3 Key Design Decisions

I assemble in COO and convert to CSR once, never through a dense matrix. The physics (a, f, g) is isolated in `model.py` so the same assembler runs P4, P5 and the layered jump. The default solve is sparse direct: at the meshes in this report ($n \le 254$, $N_{\text{unk}} \le 6.5 \times 10^4$, five nonzeros per row) a banded factorisation is cheaper than unpreconditioned CG, which I still run as a residual-stopped iterative check and as a contrast-sensitivity probe. The cost measure is named before it is quoted: for CG it is the iteration count to a stated relative residual; for the direct solver it is one sparse factorisation, with wall time reported only as a secondary observation. I do not count assembly, eigenvalue estimates, or plotting.

4 Verification and Numerical Experiments

4.1 Verification: Method of Manufactured Solutions

The bank's finite-difference self-check on $f = -\nabla\cdot(a\nabla u^*)$ passes with worst relative error 5.8×10^{-6}. That checks the analytic data, not my solver [8]. Sign and boundary gates come first: P2 ($u^* = x^2 - y^2$) has $L^\infty = 3.3 \times 10^{-16}$ at $n = 15$ (the five-point operator is exact on this quadratic); P5 recovers the Tutorial 5 centre $u_h(0.5, 0.5) = 1.652$.

Table 1 is the P4 refinement study. The named norms are the discrete L^2 norm $h\|u_h - u^*\|_2$ and $\|u_h - u^*\|_\infty$. Both fitted log-log slopes on the four uncontaminated levels $n = 15, 31, 63, 127$ are 2.00 (standard error of the fit $< 10^{-3}$; max log-deviation 0.000). Figure 3 is that table as a plot; Figure 1 shows the field and the mesh convention; Figure 2 is the sparsity pattern of the eliminated matrix.

Table 1: P4 smooth manufactured solution, face coefficient at the half-point, `spsolve`. Discrete L^2 means $h\|u_h - u^*\|_2$.

n	h	L^2	L^∞	$\|b - Au_h\|_\infty$	p_{L^2}	p_∞
15	1/16	1.598×10^{-3}	3.205×10^{-3}	4.48×10^{-13}	—	—
31	1/32	3.988×10^{-4}	8.009×10^{-4}	3.00×10^{-12}	2.00	2.00
63	1/64	9.966×10^{-5}	2.003×10^{-4}	1.40×10^{-11}	2.00	2.00
127	1/128	2.491×10^{-5}	5.007×10^{-5}	6.92×10^{-11}	2.00	2.00

Table 2: P4 truncation $h\|b - A_h u^*\|_2$. Each refinement multiplies h by 1/2 and the truncation by $\approx 1/4$.

n	h	$h\|\tau_h\|_2$	$\|\tau_h\|_\infty$
15	1/16	4.978×10^{-2}	1.028×10^{-1}
31	1/32	1.248×10^{-2}	2.572×10^{-2}
63	1/64	3.122×10^{-3}	6.433×10^{-3}
127	1/128	7.808×10^{-4}	1.609×10^{-3}

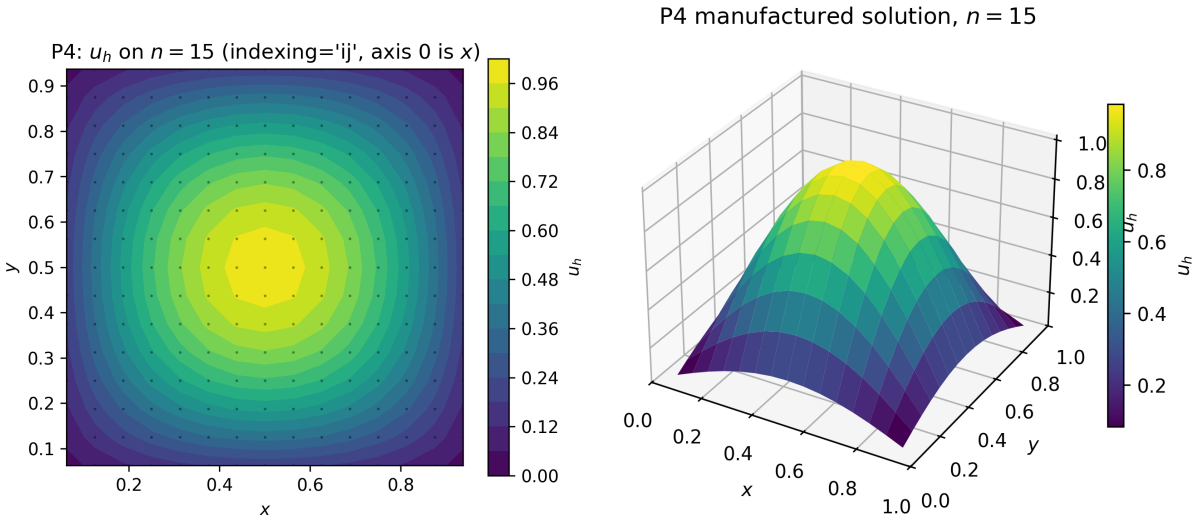


Figure 1: P4 solution at $n = 15$. Left: filled contours with interior nodes overplotted; `indexing='ij'`, axis 0 is x. Right: surface. The figure exists to show what the mesh actually resolved, not as a convergence claim.

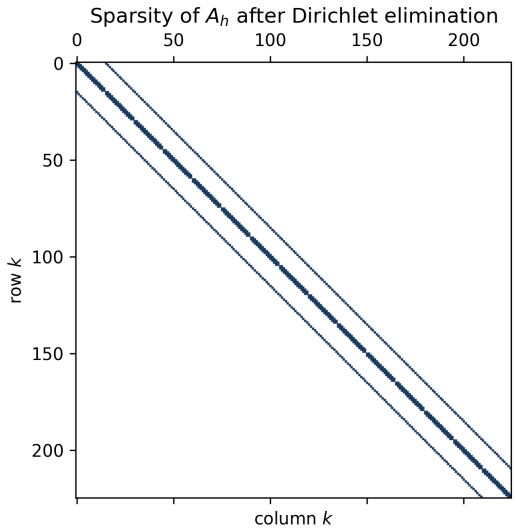


Figure 2: Spy plot of A_h at $n = 15$ after Dirichlet elimination (225×225, five-point bandwidth). The pattern is evidence of the index map $k = in + j$, not decoration.

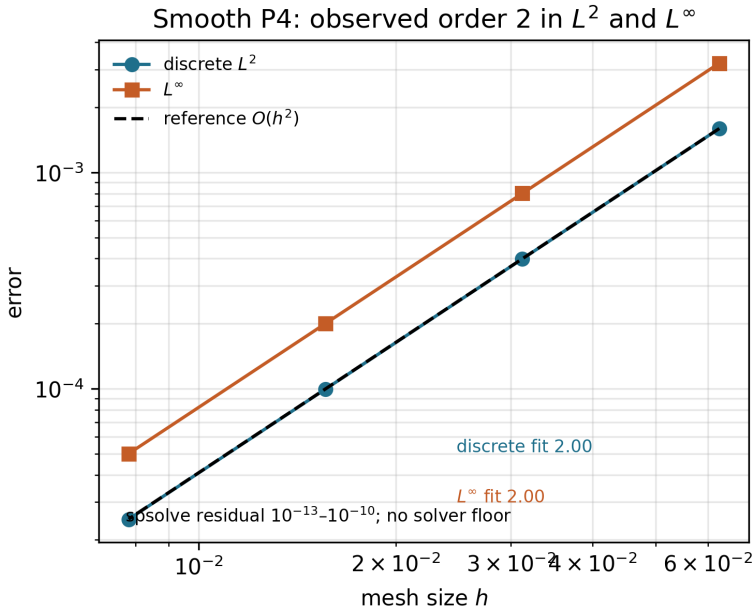


Figure 3: Log-log P4 errors versus h. Fitted discrete-L^2 order 2.001 ± 0.000, L^∞ order 2.000 ± 0.000, four uncontaminated levels, reference slope 2. This is the headline verification that the face-averaged stencil is second order on smooth data.

An independent 2D P1 FEM solve of the same P4 data (`scikit-fem`, homogeneous Dirichlet by `condense`) gives $L^2 = 5.18 \times 10^{-3}$ at $h_{\max} = 8.84 \times 10^{-2}$, residual 2×10^{-15}. That is a different mesh and a different method; it is an order-of-magnitude oracle, not a digit-for-digit check. The 1D P1 training problem $-u'' = \pi^2 \sin(\pi x)$ with nodal trapezoidal load fits discrete-L^2 order 2.00; the hand-checked element matrix is

$$K^{(e)} = \frac{1}{h}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad h = \tfrac{1}{4} \Rightarrow \begin{pmatrix} 4 & -4 \\ -4 & 4 \end{pmatrix}.$$

4.2 Separating Algebraic and Discretisation Error

`spsolve` has no tolerance parameter. I treat its solution as the exact discrete solution and report the backward residual. On P4 at $n = 31$, $\|b - Au_h\|_\infty = 3.00 \times 10^{-12}$ while the discrete L^2 error is 3.99×10^{-4}. A crude forward-error sketch $\|e\|_2 \lesssim \|r\|_2/\lambda_{\min}$ with $\lambda_{\min} \approx 28.7$ puts the algebraic error near 10^{-13}, four orders below the discretisation error.

Unpreconditioned CG on the same matrix, stopped on the relative residual $\|r_k\|_\infty/\|r_0\|_\infty$, is the second demonstration (Figure 4). At relative tolerance 10^{-2} the discrete L^2 error is still 4.33×10^{-4}; from 10^{-4} onwards it has locked onto the direct value 3.988×10^{-4} (Table 3). Tightening further does not change the reported discretisation error, so the fitted order in Table 1 is not a solver artefact. I stop CG on the residual, never on $\|x^{k+1} - x^k\|$.

Table 3: P4, $n = 31$: CG versus `spsolve`. Cost for CG is iteration count; `spsolve` is one factorisation (0.002 s wall). The L^2 column is the quantity of interest.

Solve	iterations	$\|b - Au\|_\infty$	discrete L^2
CG, rel. 10^{-2}	26	7.5×10^{-1}	4.329×10^{-4}
CG, rel. 10^{-4}	39	5.9×10^{-3}	3.988×10^{-4}
CG, rel. 10^{-6}	43	1.3×10^{-5}	3.988×10^{-4}
CG, rel. 10^{-10}	47	3.5×10^{-9}	3.988×10^{-4}
`spsolve`	—	3.0×10^{-12}	3.988×10^{-4}

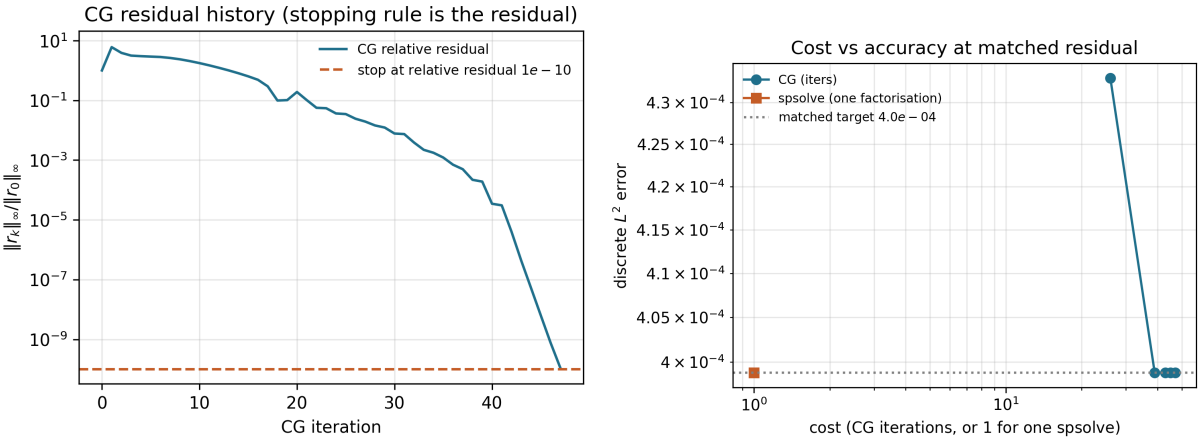


Figure 4: Left: CG relative residual versus iteration, with the stopping line at 10^{-10}. Right: discrete L^2 error versus CG iterations; the dashed line is the `spsolve` error. Once the residual is tight enough the error does not fall—that floor is discretisation, not a better solver. At $n = 31$ there is no runtime crossover: both solves finish in about 2 ms.

4.3 Depth Comparison: Harmonic versus Arithmetic Face Coefficients

The comparison uses the same layered problem, the same odd-N grids, the same Dirichlet data (3), and the same direct solver. The only change is the face rule at every edge, including the interface. Table 4 and Figures 5–6 are the evidence.

Harmonic averaging is nodally exact: $L^\infty \le 3 \times 10^{-14}$ on all four meshes, and the numerical flux $a_E(U_R - U_L)/h$ equals 20/11 to at most 5×10^{-13}. That is the pack's sanity check [8]. An interface-fitted 1D P1 FEM with 32 elements (so $x = 1/2$ is a node, the opposite alignment of the FDM grid) reproduces the same profile to $L^\infty = 6.7 \times 10^{-16}$. Two methods that share neither the stencil nor the mesh convention agree on the exact nodal values.

Arithmetic averaging is a consistent elliptic operator and still produces a symmetric M-matrix—and it is the wrong interface resistance. The flux jump stays ≈ 0.66 under refinement (it does not converge to zero). The L^∞ solution error is first-order ($\approx 1.94 \times 10^{-2}$ down to 2.38×10^{-3} as N runs through $31, 63, 127, 255$). A symmetric matrix alone does not prove that the interface physics is correct.

On every harmonic and arithmetic run the discrete solution stays in $[0, 1]$. I read that as the M-matrix sign pattern, not as an accuracy claim: the arithmetic solution is equally monotone and equally wrong at the jump.

Table 4: Layered $a = 1 \,|\, 10$, odd interval counts so $x = 1/2$ bisects a grid edge. Flux jump is $\max_j |a_E(U_R - U_L)/h - 20/11|$.

n	N	harm. L^∞	harm. flux jump	arith. L^∞	arith. flux jump
30	31	3.8×10^{-15}	3.3×10^{-14}	1.94×10^{-2}	0.663
62	63	8.9×10^{-15}	1.3×10^{-13}	9.60×10^{-3}	0.662
126	127	2.5×10^{-14}	1.6×10^{-13}	4.78×10^{-3}	0.662
254	255	2.7×10^{-14}	5.2×10^{-13}	2.38×10^{-3}	0.662

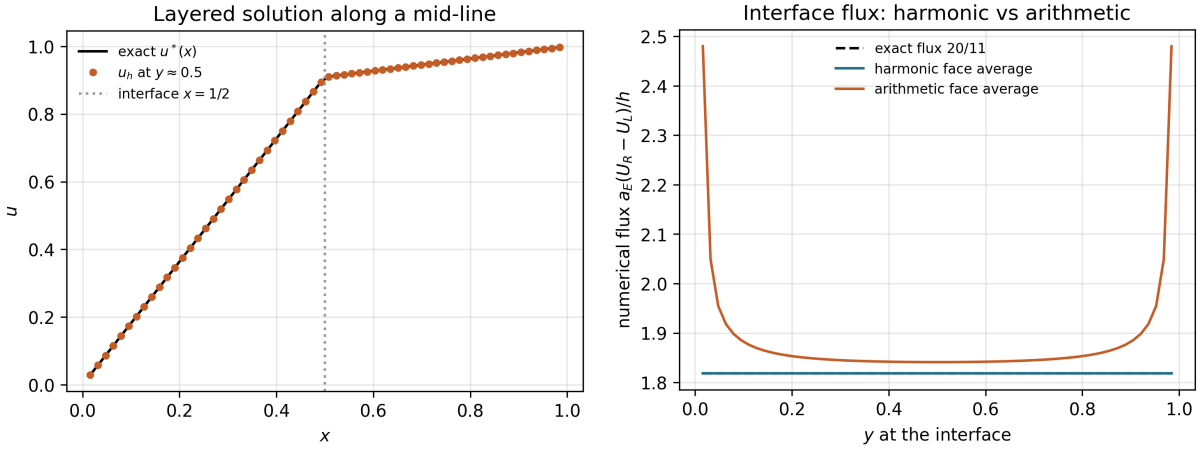


Figure 5: Left: harmonic u_h along $y \approx 0.5$ against the exact piecewise-linear profile; the kink at $x = 1/2$ is resolved by placing the interface between columns. Right: numerical interface flux versus y. Harmonic sits on $20/11$; arithmetic is a different constant. That gap is the wrong resistance, and it does not shrink with h.

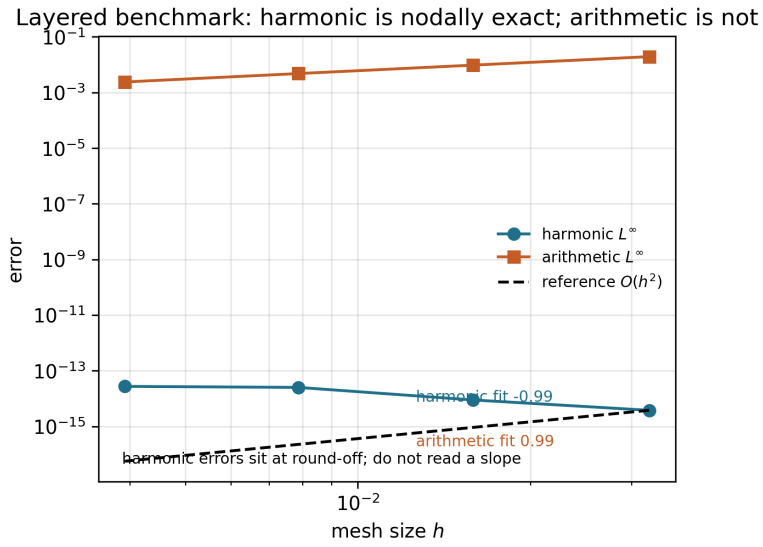


Figure 6: Layered L^∞ versus h. Harmonic errors are round-off (do not fit a slope). Arithmetic tracks first order. The figure is the depth comparison: two face rules, matched data, named norm.

4.4 Main Results: Contrast, Conditioning, Cost

Unpreconditioned CG iteration counts on the layered harmonic matrix at $n = 30$, relative residual 10^{-10}, grow with the contrast a_R/a_L: 125, 227, 401, 615 iterations at contrasts $2, 10, 50, 200$ (Figure 7). Coefficient contrast worsens conditioning but does not, by itself, break the discrete maximum principle. On the smooth P4 problem, $\kappa_2(A_h)$ tracks h^{-2} as already quoted. At the meshes I actually solved, `spsolve` remains the cheaper way to reach a residual below the discretisation error; I did not observe a CG–direct crossover for $n \leq 31$.

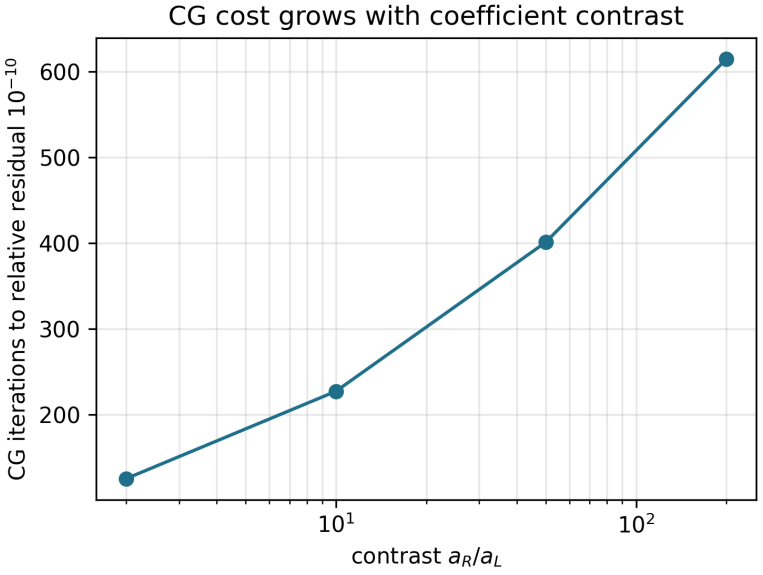


Figure 7: CG iterations versus conductivity contrast on the layered problem, $n = 30$, residual stop 10^{-10}. The cost measure here is iterations, not wall time. Contrast is expensive for unpreconditioned CG; it is not a reason to switch to an arithmetic face average.

4.5 Limitations and Robustness

The second-order claim is for smooth a and the half-point (or equivalent arithmetic-on-smooth) face rule. Across a jump, harmonic averaging on an aligned grid is exact for this piecewise-linear u^*; that is a 1D identity, not a general theorem for rough interfaces that cut through cells at arbitrary angles. Arithmetic averaging remains an M-matrix and will still look "stable" on a colour plot. I would not use it as a reference closure.

I did not implement algebraic multigrid on the FDM matrix, nor a 2D interface-fitted FEM comparison beyond the single P4 oracle. Unpreconditioned CG will not be the iterative method of choice on finer 2D meshes; the Tutorial 6 AMG-CG demo is the natural next solver, not a result of this report. The scheme is two-dimensional and Cartesian; a cut-cell disk would need a different boundary row.

5 Conclusions

- I assembled a conservative face-averaged five-point scheme for $-\nabla \cdot (a\nabla u)$ with Dirichlet data moved into b, and I verified it on the analytic bank before the jump.
- On smooth P4, discrete L^2 and L^∞ both fit order 2.00 over four meshes; $\|A - A^\top\|_{\max} = 0$ and $\lambda_{\min} > 0$. The reported rate survives a CG tolerance sweep.
- On the layered jump, harmonic face coefficients are nodally exact and keep the flux at 20/11. Arithmetic coefficients give a first-order solution error and an $\mathcal{O}(1)$ flux jump. Symmetry is necessary and not sufficient.
- Cost is CG iterations or one sparse factorisation. At $n = 31$, CG and `spsolve` agree in L^2 once the relative residual is 10^{-4}; there is no runtime crossover on this range. CG iterations grow with contrast.

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A AI Transparency Log

A.1 A1. Tools used

Tool	Date	Primary use
Cursor (Grok 4.6 agent)	13–14 Oct 2026	Week 5 P5 assembly, Dirichlet shift, issue ticket
Cursor (Grok 4.6 agent)	20 Oct 2026	Week 6 P4 MMS table and log-log slope
Cursor (Grok 4.6 agent)	22 Oct 2026	Layered harmonic/arithmetic comparison, `run_all.py`, this report

A.2 A2–A3. Claim audits

Item 1: conservative face stencil and Dirichlet sign. Prompt/claim: assemble $-\nabla \cdot (a\nabla u)$ with $b_k \mathrel{+}= a_{\text{face}} g/h^2$. Risk: the wrong sign on the boundary shift (or treating boundary nodes as unknowns) still produces a smooth-looking u_h. Independent check: P5 centre 1.651995 against the bank value 1.652 at $n = 15$; P2 $L^\infty = 3.3 \times 10^{-16}$; hand matrix $n = 2$ has diagonal $4/h^2 = 36$ and nnz = 12 (`validation_tests.py`). Decision: keep the interior-only unknown list; reject any row-replacement Dirichlet that leaves columns in place.

Item 2: layered exact profile. Prompt/claim: an extracted PDF line looked like $u^* = 2x+9/11$ on the right half. Risk: that formula is discontinuous at $x = 1/2$ (10/11 versus 20/11) and would falsify every "nodally exact" test. Independent check: I re-derived u^* from continuity of u and of au_x with $u(0) = 0$, $u(1) = 1$, obtaining (3). Harmonic FDM then hits $L^\infty \sim 10^{-14}$; the wrong formula cannot. Decision: use (3); do not trust OCR of the problem pack for the piecewise definition.

Item 3: arithmetic averaging as a "symmetric, hence correct" scheme. Prompt/claim: any consistent positive face average yields a valid interface scheme. Risk: reporting $\|A - A^\top\| = 0$ as evidence that the jump is right. Independent check: Table 4—arithmetic is symmetric and monotone, yet the flux jump stays 0.66 and L^∞ is only first order. Harmonic is nodally exact. Decision: quote symmetry as a structure fact; quote flux and nodal error as the physics facts.

A.3 A4. Ownership

I own the stencil derivation, the layered exact profile, the choice of odd-N grids, the interpretation of arithmetic versus harmonic, and the figures. AI assistance filled and checked `assemble.py` / `run_all.py` / `validation_tests.py` and the first typesetting of this file. Trusted dependencies: `scipy.sparse.coo_matrix` and `spsolve` for the discrete solve; `scipy.sparse.linalg.cg` with a residual callback; `eigsh` only as a numerical SPD/conditioning corroboration; `scikit-fem` `condense`/`asm` as an optional 2D oracle. I can explain and modify the FDM assembler without AI during the viva.

B Individual Contribution Statement

This is a solo project, so there is one ICS.

Individual Contribution Statement — Project 2 (PDE BVP)

Name: Chuyang Wang Student number: 2363591

Role this project: sole author (implementation, weekly tasks, report).

The specific things I did this project:

1. Derived the conservative stencil (5) and the Dirichlet shift (7); implemented them in `code/assemble.py` with the index map $k = in + j$.

2. Completed the Week 5 P5 challenge ($u_h(0.5, 0.5) = 1.652$) and the Week 6 four-mesh slope (2.00 in discrete L^2 and L^∞).

3. Re-derived the layered profile (3), implemented harmonic versus arithmetic face averages, and produced Table 4 and Figures 5–6.

4. Separated algebraic from discretisation error by a CG residual sweep (Table 3); measured CG iterations against contrast (Figure 7).

5. Wrote `validation_tests.py` and `run_all.py`, and wrote the full LaTeX report including the AI log and this ICS.

What I would do differently: freeze the layered exact formula on paper before reading the OCR'd pack, instead of debugging a "failed" nodal-exact test against a discontinuous u^*. Anything I should have asked for help with earlier: whether odd N (interface between columns) versus even N (interface on a node) is required for the FDM sanity check, before running the first layered table on $n = 15$. Signature: _____ Date: _____

C Supplementary data

P4 structure at $n = 15$: $\|A - A^\top\|_{\max}/\|A\|_{\max} = 0$, $\lambda_{\min} = 28.61$, $\kappa_2 \approx 1.27 \times 10^2$.
FEM 1D element matrix for $h = 1/4$: $\left(\begin{smallmatrix} 4 & -4 \\ -4 & 4 \end{smallmatrix}\right)$.
All numbers in the tables are reproduced by `python code/run_all.py`.